PRAKTIKUM BASIS DATA

MODUL 03

(Scripting SQL Basic)

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

TP

1. **Soal 1**
   Jawaban:
   CREATE TABLE Mahasiswa (
   id number primary key,
   nama varchar2 (50),
   tempat_lahir varchar2 (50),
   tanggal_lahir date,
   nomor_hp varchar2 (15),
   email varchar2(50),
   berat_badan number,
   tinggi_badan number
   );

```
CREATE TABLE Mahasiswa (
    id number primary key,
    nama varchar2 (50),
    tempat_lahir varchar2 (50),
    tanggal_lahir date,
    nomor_hp varchar2 (15),
    email varchar2(50),
    berat_badan number,
    tinggi_badan number
);
```

2. **Soal 2**
   Jawaban:
   insert into Mahasiswa values (1, 'titu', 'surian', to_date('23-08-06','dd-mm-yy'), '081378801993', 'titu@gmail.com', 45, 165);
   insert into Mahasiswa values (2, 'quratu', 'solok', to_date('03-06-05','dd-mm-yy'), '081378805553', 'quratu@gmail.com', 40, 160);
   insert into Mahasiswa values (3, 'ayun', 'padang', to_date('21-08-06','dd-mm-yy'), '081376543993', 'ayun@gmail.com', 50, 140);
   insert into Mahasiswa values (4, 'defaren', 'surian', to_date('29-09-06','dd-mm-yy'), '081398765432', 'defaren@gmail.com', 60, 170);

insert into Mahasiswa values (5, 'farah', 'surian', to_date('29-09-07','dd-mm-yy'), '08139822232', 'farah@gmail.com', 65, 175);
insert into Mahasiswa values (6, 'dibba', 'medan', to_date('12-04-06','dd-mm-yy'), '081899965432', 'dibba@gmail.com', 69, 180);
insert into Mahasiswa values (7, 'vella', 'surian', to_date('20-10-98','dd-mm-yy'), '081398333332', 'vella@gmail.com', 65, 160);
insert into Mahasiswa values (8, 'zaki', 'surian', to_date('26-08-05','dd-mm-yy'), '081300995432', 'zaki@gmail.com', 55, 155);
insert into Mahasiswa values (9, 'zahran', 'padang', to_date('25-12-05','dd-mm-yy'), '081998765432', 'zahran@gmail.com', 69, 171);
insert into Mahasiswa values (10, 'zea', 'solok', to_date('12-09-11','dd-mm-yy'), '081396666432', 'zea@gmail.com', 60, 159);
insert into Mahasiswa values (11, 'nabila', 'surian', to_date('11-09-05','dd-mm-yy'), '081398700002', 'nabila@gmail.com', 60, 170);
insert into Mahasiswa values (12, 'asta', 'jepang', to_date('30-03-10','dd-mm-yy'), '081396767432', 'Asta@gmail.com', 70, 188);
insert into Mahasiswa values (13, 'gojo', 'shibuya', to_date('28-02-04','dd-mm-yy'), '08139871132', 'gojo@gmail.com', 60, 167);
insert into Mahasiswa values (14, 'haku', 'jepang', to_date('11-03-03','dd-mm-yy'), '08139112232', 'haku@gmail.com', 50, 150);
insert into Mahasiswa values (15, 'howl', 'jepang', to_date('02-01-02','dd-mm-yy'), '08139879898', 'howl@gmail.com', 60,

ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NOMOR_HP	EMAIL	BERAT_BADAN	TINGGI_BADAN
1	titu	surian	23-AUG-06	081378801993	titu@gmail.com	45	165
2	quratu	solok	03-JUN-05	081378805553	quratu@gmail.com	40	160
3	ayun	padang	21-AUG-06	081376543993	ayun@gmail.com	50	140
4	defaren	surian	29-SEP-06	081398765432	defaren@gmail.com	60	170
5	farah	surian	29-SEP-07	08139822232	farah@gmail.com	65	175
6	dibba	medan	12-APR-06	081899965432	dibba@gmail.com	69	180
7	vella	surian	20-OCT-98	081398333332	vella@gmail.com	65	160
8	zaki	surian	26-AUG-05	081300995432	zaki@gmail.com	55	155
9	zahran	padang	25-DEC-05	081998765432	zahran@gmail.com	69	171
10	zea	solok	12-SEP-11	081396666432	zea@gmail.com	60	159
11	nabila	surian	11-SEP-05	081398700002	nabila@gmail.com	60	170
12	asta	jepang	30-MAR-10	081396767432	Asta@gmail.com	70	188
13	gojo	shibuya	28-FEB-04	08139871132	gojo@gmail.com	60	167
14	haku	jepang	11-MAR-03	08139112232	haku@gmail.com	50	150
15	howl	jepang	02-JAN-02	08139879898	howl@gmail.com	60	170

3. **Soal no 3**
   Jawaban:

Script Output × Query Result ×

SQL | All Rows Fetched: 15 in 0.004 seconds

	huruf kecil	kapital di awal	gabung	jumlah huruf	potong nama	lpad nama	rpad nama	contoh LTRIM	contoh RTRIM	posisi at	email baru
1	itu	Titu	titu Mahasiswa	4	tit	******titu	titu******	DATABASE	DATABASE	5	titu@student.com
2	uratu	Quratu	quratu Mahasiswa	6	qur	****quratu	quratu****	DATABASE	DATABASE	7	quratu@student.com
3	yun	Ayun	ayun Mahasiswa	4	ayu	******ayun	ayun******	DATABASE	DATABASE	5	ayun@student.com
4	efaren	Defaren	defaren Mahasiswa	7	def	***defaren	defaren***	DATABASE	DATABASE	8	defaren@student.com
5	arah	Farah	farah Mahasiswa	5	far	*****farah	farah*****	DATABASE	DATABASE	6	farah@student.com
6	ibba	Dibba	dibba Mahasiswa	5	dib	*****dibba	dibba*****	DATABASE	DATABASE	6	dibba@student.com
7	ella	Vella	vella Mahasiswa	5	vel	*****vella	vella*****	DATABASE	DATABASE	6	vella@student.com
8	aki	Zaki	zaki Mahasiswa	4	zak	******zaki	zaki******	DATABASE	DATABASE	5	zaki@student.com
9	ahran	Zahran	zahran Mahasiswa	6	zah	****zahran	zahran****	DATABASE	DATABASE	7	zahran@student.com
10	ea	Zea	zea Mahasiswa	3	zea	*******zea	zea*******	DATABASE	DATABASE	4	zea@student.com
11	abila	Nabila	nabila Mahasiswa	6	nab	****nabila	nabila****	DATABASE	DATABASE	7	nabila@student.com
12	sta	Asta	asta Mahasiswa	4	ast	******asta	asta******	DATABASE	DATABASE	5	Asta@student.com
13	ojo	Gojo	gojo Mahasiswa	4	goj	******gojo	gojo******	DATABASE	DATABASE	5	gojo@student.com
14	aku	Haku	haku Mahasiswa	4	hak	******haku	haku******	DATABASE	DATABASE	5	haku@student.com
15	owl	Howl	howl Mahasiswa	4	how	******howl	howl******	DATABASE	DATABASE	5	howl@student.com

a.

```
select nama,
upper(nama) as "huruf besar",
lower(nama) as "huruf kecil ",
initcap (nama) as "kapital di awal",
concat (nama,' Mahasiswa') as "gabung",
length (nama) as "jumlah huruf",
substr (nama,1,3) as "potong nama",
lpad(nama,10,'*') as "lpad nama",
rpad(nama,10,'*') as "rpad nama",
ltrim('   DATABASE') as "contoh LTRIM",
rtrim('DATABASE   ') as "contoh RTRIM",
instr (email,'@') as "posisi at",
replace (email,'gmail','student') as "email baru"
from mahasiswa;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 15 in 0.004 seconds

	R	hari	bulan	nama bulan	tahun	tanggal sekarang	tambah 6 bulan	akhir bulan	senin berikut	selisih bulan	pembulatan tahun
1		23	08	august	2006	11-MAR-26	23-FEB-07	31-AUG-06	28-AUG-06	234.6443757467144563918757467144563918 76	01-JAN-07
2		03	06	june	2005	11-MAR-26	03-DEC-05	30-JUN-05	06-JUN-05	249.2895370370370370370370370370370370 37	01-JAN-05
3		21	08	august	2006	11-MAR-26	21-FEB-07	31-AUG-06	28-AUG-06	234.7088918757467144563918757467144563 92	01-JAN-07
4		29	09	september	2006	11-MAR-26	29-MAR-07	30-SEP-06	02-OCT-06	233.4508273596176821983273596176821983 27	01-JAN-07
5		29	09	september	2007	11-MAR-26	29-MAR-08	30-SEP-07	01-OCT-07	221.4508273596176821983273596176821983 27	01-JAN-08
6		12	04	april	2006	11-MAR-26	12-OCT-06	30-APR-06	17-APR-06	238.9992144563918757467144563918757467 14	01-JAN-06
7		20	10	october	2098	11-MAR-26	20-APR-99	31-OCT-98	27-OCT-98	-871.2588500597371565113500597371565113 5	01-JAN-99
8		26	08	august	2005	11-MAR-26	26-FEB-06	31-AUG-05	29-AUG-05	246.5476015531660692951015531660692951 02	01-JAN-06
9		25	12	december	2005	11-MAR-26	25-JUN-06	31-DEC-05	26-DEC-05	242.5798596176821983273596176821983273 6	01-JAN-06
10		12	09	september	2011	11-MAR-26	12-MAR-12	30-SEP-11	19-SEP-11	173.9992144563918757467144563918757467 14	01-JAN-12
11		11	09	september	2005	11-MAR-26	11-MAR-06	30-SEP-05	12-SEP-05	246	01-JAN-06
12		30	03	march	2010	11-MAR-26	30-SEP-10	31-MAR-10	05-APR-10	191.4185692951015531660692951015531660 69	01-JAN-10
13		28	02	february	2004	11-MAR-26	28-AUG-04	29-FEB-04	01-MAR-04	264.4830854241338112305854241338112305 85	01-JAN-04
14		11	03	march	2003	11-MAR-26	11-SEP-03	31-MAR-03	17-MAR-03	276	01-JAN-03
15		02	01	january	2002	11-MAR-26	02-JUL-02	31-JAN-02	07-JAN-02	290.3217951015531660692951015531660692 95	01-JAN-02

b.

```
select nama,
```

```
tanggal_lahir,
to_char(tanggal_lahir,'dd') as "hari",
to_char(tanggal_lahir,'mm') as "bulan",
to_char(tanggal_lahir,'month') as "nama bulan",
to_char(tanggal_lahir,'yyyy') as "tahun",
sysdate as "tanggal sekarang",
add_months(tanggal_lahir,6) as "tambah 6 bulan",
last_day(tanggal_lahir) as "akhir bulan",
next_day(tanggal_lahir,'monday') as "senin berikut",
months_between(sysdate,tanggal_lahir) as "selisih bulan",
round(tanggal_lahir,'year') as "pembulatan tahun"
from mahasiswa;
```

Script Output x | Query Result x

SQL | All Rows Fetched: 15 in 0.005 seconds

	NAMA	tanggal format	tinggi char	contoh number	contoh date
1	titu	23-08-2006	165	123	01-JAN-24
2	quratu	03-06-2005	160	123	01-JAN-24
3	ayun	21-08-2006	140	123	01-JAN-24
4	defaren	29-09-2006	170	123	01-JAN-24
5	farah	29-09-2007	175	123	01-JAN-24
6	dibba	12-04-2006	180	123	01-JAN-24
7	vella	20-10-2098	160	123	01-JAN-24
8	zaki	26-08-2005	155	123	01-JAN-24
9	zahran	25-12-2005	171	123	01-JAN-24
10	zea	12-09-2011	159	123	01-JAN-24
11	nabila	11-09-2005	170	123	01-JAN-24
12	asta	30-03-2010	188	123	01-JAN-24
13	gojo	28-02-2004	167	123	01-JAN-24
14	haku	11-03-2003	150	123	01-JAN-24
15	howl	02-01-2002	170	123	01-JAN-24

c.

```
select nama,
to_char(tanggal_lahir,'DD-MM-YYYY') as "tanggal format",
```

```
to_char(tinggi_badan,'999') as "tinggi char",
to_number('123') as "contoh number",
to_date('2024-01-01','YYYY-MM-DD') as "contoh date"
from Mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.003 seconds

	NAMA	selisih berat	sisa bagi	floor tinggi	pangkat dua	pembulatan	pembulatan_atas
1	titu	25	0	55	27225	15	15
2	quratu	30	0	53	25600	13.33	14
3	ayun	20	0	46	19600	16.67	17
4	defaren	10	0	56	28900	20	20
5	farah	5	0	58	30625	21.67	22
6	dibba	1	0	60	32400	23	23
7	vella	5	0	53	25600	21.67	22
8	zaki	15	0	51	24025	18.33	19
9	zahran	1	1	57	29241	23	23
10	zea	10	4	53	25281	20	20
11	nabila	10	0	56	28900	20	20
12	asta	0	3	62	35344	23.33	24
13	gojo	10	2	55	27889	20	20
14	haku	20	0	50	22500	16.67	17
15	howl	10	0	56	28900	20	20

d.

```
select nama,
abs(berat_badan-70) as "selisih berat",
mod(tinggi_badan,5) as "sisa bagi",
floor(tinggi_badan/3) as "floor tinggi",
power(tinggi_badan,2) as "pangkat dua",
round(berat_badan/3,2) as "pembulatan",
ceil(berat_badan/3) as "pembulatan_atas"
from mahasiswa;

SELECT * FROM Mahasiswa;
```